Technical Specification: Unified Time Attendance System (UTAS)

System Architecture & Guide for Connecting Multiple Companies

Version 3.2 - Simplified Technical Reference

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1. System Overview

This document describes a software solution designed to bring together different types of time attendance machines into one single database capable of managing multiple companies at once. The system fixes the problem of old machines (using the old Cable/UDP method) not being able to talk to new machines (using the new Internet/HTTP method).

Primary Objectives:

1. **Unified Communication:** Handling both "Push" (New style) and "Pull" (Old style) data at the same time.
2. **Universal Support:** Works with almost any ZKTeco machine made from 2008 to today.
3. **Multi-Company Support:** Keeps data strictly separate for different companies using the same server.
4. **Security:** Uses encryption and strict rules to keep data safe.

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2. System Architecture

2.1 Hybrid Data Collection Engine

The software runs two separate services side-by-side to handle the different languages the machines speak.

Service A: The "Push" Receiver (ADMS/HTTP)

- **Method:** Web / Internet (HTTP POST).
- **Function:** Acts like a website that receives real-time data from machines.
- **Capacity:** Can handle over 5,000 machines sending data at the same time.
- **Data Flow:** Machine -> Internet -> Server.

Service B: The "Pull" Service (Legacy SDK)

- **Method:** Network Cable (UDP Port 4370).
- **Function:** A background task that actively goes out and fetches data from specific machines.
- **How it works:**
    1. **Connect:** Opens a connection to the machine's IP address.
    2. **Login:** Uses a password (Comm Key) if set.
    3. **Download:** Asks for all new logs created since the last time it checked.

4. **Disconnect:** Closes the connection.

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3. Hardware Support List

We group machines into three levels based on how they communicate.

Level 1: Upgradeable Legacy Hardware

Machines made between 2011-2018. These often look like they don't work, but they actually have a "hidden" feature (ADMS) that we can turn on.

Series	Model Examples	What is needed?
iClock	260, 360, 480, 580, 680, 880	Change settings in menu.
K-Series	K14, K20, K40, K50, K60	K14/K40 might need a software update.
S-Series	S500, S560, S922 (Portable)	Works over Wi-Fi/Sim Card.
F-Series	F18, F19, F21, F22	Common for door access.
Standard	IN01, UA300, MB20, VF380	Ready to go.

Level 2: Strictly Old Hardware (Pull Only)

Very old machines (pre-2011) that **cannot** use the modern Internet method. They must be managed by our **"Pull" Service**.

- **X-Series:** X628 (Black & White Screen), older X628-C.
- **Old Access Control:** MA300, MA500.
- **No-Name Brands:** Unbranded machines using ZKTeco parts.

Level 3: Modern Hardware

New machines that work perfectly out of the box.

- **Green Label:** G1, G2, G3.
- **Visible Light:** SpeedFace V3L, V4L, V5L.

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4. Multi-Company Architecture

4.1 Data Routing (Who owns this data?)

The system strictly separates data so one company never sees another's.

- **Identifier:** We use the Machine's Serial Number (SN).
- **The Rule:** A central "Map" decides where data goes.
    - `Serial No 888` -> Belongs to -> `Company A`
    - `Serial No 999` -> Belongs to -> `Company B`

4.2 Database Separation

We can store the data in two ways:

- **Option A: Shared Database (Simpler)**
    - One single database for everyone.
    - Every row of data has a `company_id` tag.
    - The software automatically hides Company A's data when Company B is logged in.
- **Option B: Separate Databases (More Secure)**
    - Totally internal databases for each company.
    - The system connects to `Database A` or `Database B` depending on who the machine belongs to.

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5. Security & Deployment

5.1 Internet Safety

Putting a server on the public internet requires care.

Secure Method for Modern Machines (HTTPS)

- **Encryption:** Use SSL (like a secure banking website).
- **Gateway:** Use a standard web proxy (Nginx) to filter traffic.
- **Firewall:** Only allow web traffic (Port 443). Block everything else.

Secure Method for Old Machines (UDP)

Risk: The old machine language (Port 4370) is not secure. **NEVER** open this port to the public internet. hackers can attack it. **Solution:**

- **Method A (VPN):** Create a secure private tunnel (VPN) between the office and the server.
- **Method B (Local Collector):** Put a small computer (like a Raspberry Pi) at the office to collect data locally and send it securely to the cloud.

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6. Server Computer Requirements

The system is very efficient and does not need a powerful supercomputer.

Size of Deployment	Number of Machines	Hardware Needed
Small / Pilot	1 - 20	**Basic Cloud Server** 1 CPU, 1 GB RAM, 10 GB Disk
Standard	20 - 200	**Standard VPS** 2 CPU, 4 GB RAM, 50 GB Disk
Enterprise	200 - 2,000	**Focused Performance** 4 CPU, 8-16 GB RAM, Fast SSD Storage

Large Scale	2,000+	**Cluster** Multiple servers working together

Important: The speed of the Hard Drive (Disk Write Speed) is more important than the CPU speed, especially when thousands of people clock in at 9:00 AM.

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7. Migration Steps (How to Switch)

1. **Audit:** Find all machines and write down their IP addresses.
2. **Sort:** Decision - "Can this machine use the Internet method (Level 1) or must it use the Cable method (Level 2)?"
3. **Configure Level 1:** Change the settings in the machine menu to point to the new server IP.
4. **Configure Level 2:** Add the IP addresses of the old machines to the Server's "Pull List" (Requires VPN).
5. **Test:** Check the system to ensure data is appearing for all companies.